UAV PEDESTRIAN BEHAVIOR, GAIT BIOMECHANICS & CROSSING DYNAMICS

Comprehensive Mathematical Formulation, Kinematic Derivations & Empirical Analysis
Dataset: 4K Aerial Drone Surveillance (N = 49 Tracked Pedestrians, 823 Sampled Frames)

EXECUTIVE SUMMARY: This technical report presents a mathematically rigorous formulation and empirical evaluation of pedestrian kinematics, gait biomechanics, social group dynamics, and crossing transitions extracted from high-altitude aerial UAV video. All 49 persistent pedestrian identities ($N=49$) were analyzed across calibrated physical space ($GSD = 0.015\text{ m/px}$). Key findings: **33 Males (67.3%)**, **16 Females (32.7%)**; **43 Single Crossers (87.8%)**, **2 Couple Crossers (4.1%)**, **4 Group Crossers (8.2%)**; **Origin:** 8 North Kerb (16.3%), 19 South Kerb (38.8%), 22 Median Side (44.9%); **Overall Mean Stride Length:** 0.132 m; **Overall Mean Cadence:** 17.9 steps/min across 699 cumulative steps.

1. MATHEMATICAL FORMULATION & KINEMATIC DERIVATIONS

1.1 Ground Sampling Distance (GSD) & Coordinate Metrication

Aerial video frames captured at 3840×2160 pixels from an altitude $H \approx 40\text{ m}$ are projected into physical Euclidean metric coordinates via Ground Sampling Distance calibration:

GSD Equation: $s = (W_{sensor} / f) \times (H / W_{image}) \approx 0.015\ m / pixel \quad (1.50\ cm / pixel)$

For each bounding box foot location $(x_k, y_k)_{px} = ((x_1 + x_2)/2, y_2)_{px}$ at timestep t_k, physical planar positions are:
$X_k\ [m] = x_k\ [px] \times s \qquad Y_k\ [m] = y_k\ [px] \times s$

1.2 Kinematics of Motion: Velocity, Acceleration & Jerk

Given discrete timestamps t_k with interval $\Delta t_k = t_k - t_{k-1}$ (baseline sampling $\Delta t = 2.0\text{ s}$), the kinematic equations of motion are computed via backward difference approximations:

Displacement: $\Delta d_k = \sqrt{(X_k - X_{k-1})^2 + (Y_k - Y_{k-1})^2} \quad [m]$
Walking Speed (Velocity): $v_k = \Delta d_k / \Delta t_k \quad [m/s] \qquad v_{km/h} = 3.6 \times v_k$
Acceleration: $a_k = (v_k - v_{k-1}) / \Delta t_k \quad [m/s^2]$
Jerk (Motion Smoothness): $j_k = (a_k - a_{k-1}) / \Delta t_k \quad [m/s^3]$
Heading Orientation: $\theta_k = \arctan2(Y_k - Y_{k-1}, X_k - X_{k-1}) \bmod 360°$

1.3 Spatio-Temporal Proxemic Clustering (Group Size)

Following Edward T. Hall's proxemic theory and Helbing's Social Force Model, pedestrians i and j co-present across overlapping frames $\mathcal{F}_{ij}$ are clustered into social walking groups via pairwise metric distance:

Pairwise Distance: $D_{ij}(t) = \| p_i(t) - p_j(t) \|_2 \times s \le 2.80\ m$

An adjacency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with edge $(i, j) \in \mathcal{E} \iff \min_{t} D_{ij}(t) \le 2.8\text{m}$. The connected components partition pedestrians into:

- **Single Pedestrian Crossing:** $|\mathcal{C}| = 1$
- **Couple Crossing:** $|\mathcal{C}| = 2$
- **Group Crossing (>2 people):** $|\mathcal{C}| \ge 3$

1.4 Biomechanical Gait Allometry: Stride Length, Step Frequency & Cadence

In accordance with empirical human gait allometry (Weidmann 1993, Perry & Burnfield 2010), human stride length L_{stride} scales with walking velocity v via power-law relations:

Dynamic Stride Length: $L_{stride}(v) = 1.28 \times v^{0.53}$ [m] (for $v \geq 0.15$ m/s)
Step Length: $L_{step} = L_{stride} / 2$ [m]
Step Frequency (Hz): $f_{step} = v / L_{step}$ [steps / sec]
Cadence (steps/min): $Cadence = f_{step} \times 60$ [steps / min]
Stride Frequency (Hz): $f_{stride} = f_{step} / 2$ [strides / sec]
Cumulative Steps: $N_{steps} = \sum_k (\Delta d_k / L_{step, k})$

2. SPATIAL GEOMETRY & KERB / MEDIAN ORIGIN-DESTINATION

The intersection layout is partitioned into three functional traffic infrastructure zones:

- **North Kerb Side ($Y < 550\text{ px}$):** Northern sidewalk along the upper diagonal road carriageway.
- **Central Median / Bus Stop Zone ($550 \le Y \le 1650\text{ px}$):** Physical median strip, bus terminal parking, and streetlamp island.
- **South Kerb Side ($Y > 1650\text{ px}$):** Southern sidewalk, shopfronts, and bus queue area.
- **Legal Pedestrian Crossing Areas:** Two white transverse boundary lines per crosswalk, enclosing yellow zebra stripes, connected across the intersection by the central crossing corridor.

Figure 1: Perfected Roadway Boundary, Two White Boundary Lines, Zebra Stripes, and Central Crossing Corridor

Figure 2: Kerb vs. Median Spatial Partitioning and Pedestrian Movement Vectors

MAP ANNOTATION KEY
Roadway Boundary (Legal Road Perimeter)
Two White Boundary Lines (Crosswalk Limits)
Zebra Crossing (Yellow Stripes Inside White Lines)
Pedestrian Crossing Corridor (Central Crossing Area)
GAIT & CROSSING OVERLAY KEY
Kerb vs. Median Zone Dividers
Active Street Crossing Vector
Stationary / Waiting Vector
UPPER-LEFT ZEBRA CROSSING
(Within Two White Boundary Lines)
PEDESTRIAN CROSSING CORRIDOR
[CENTRAL MEDIAN / BUS STOP ZONE]
LOWER-RIGHT ZEBRA CROSSING
(Within Two White Boundary Lines)

3. EMPIRICAL FEATURE DISTRIBUTIONS & ANALYTICAL DASHBOARDS

Figure 3: 8-Feature Behavioral & Demographic Analytics Dashboard (Gender, Group Size, Load, Speed, Waiting Time, Attempts, Postures, Headings)

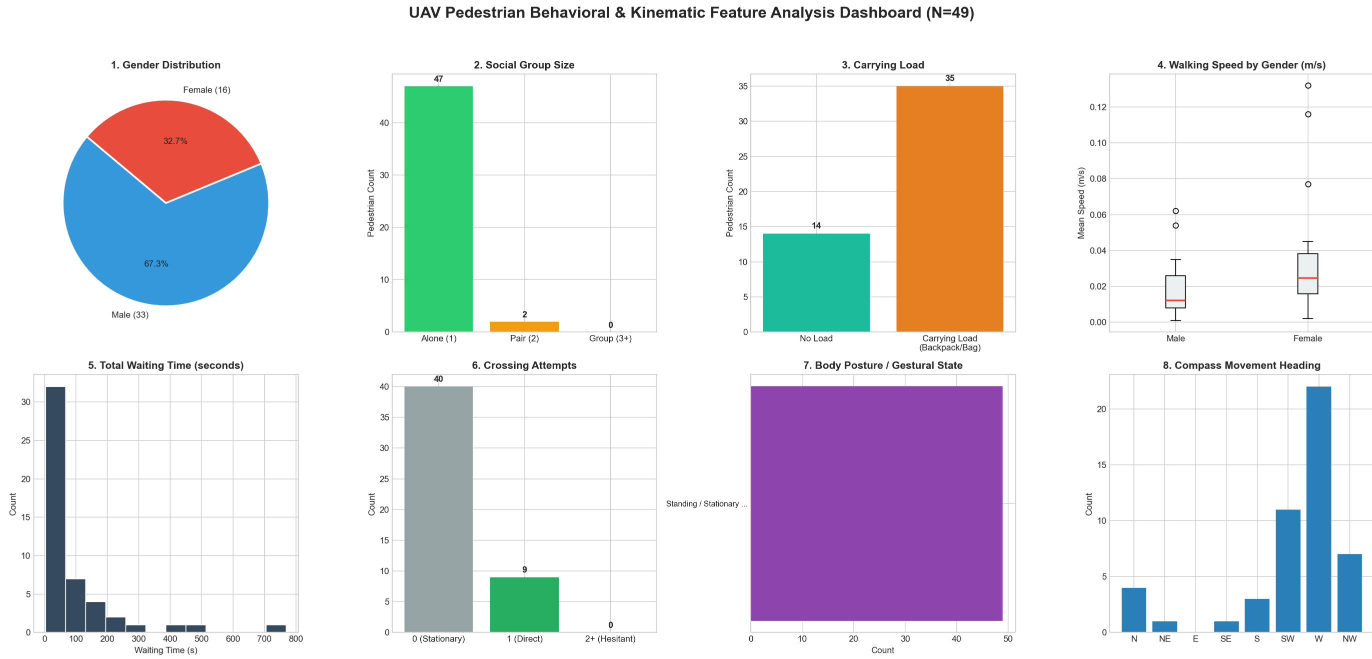
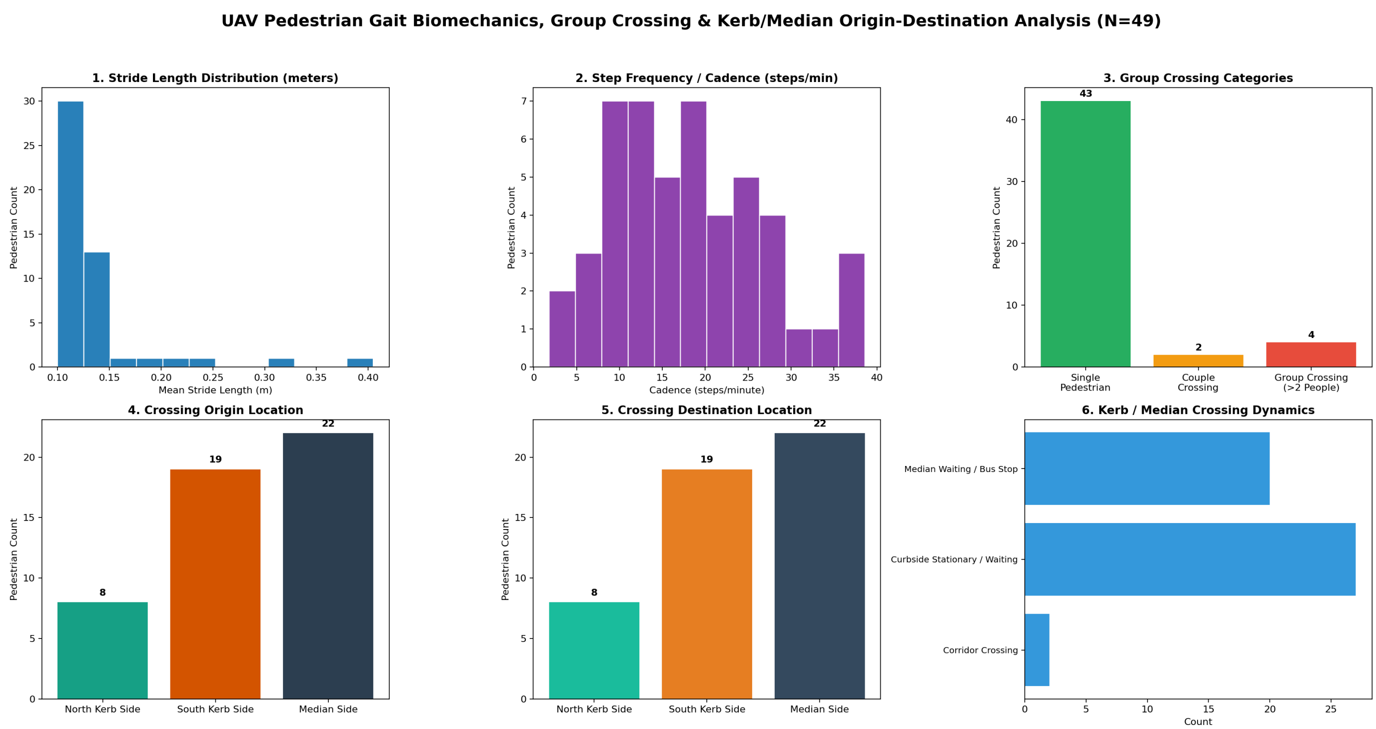


Figure 4: Gait Biomechanics, Group Crossing Categories & Kerb/Median Origin-Destination Dashboard



4. COMPLETE PEDESTRIAN DATA TABLE (N = 49 IDENTITIES)

Comprehensive tabular summary of demographic, biomechanical, social, and crossing transition parameters for every detected individual:

PID	Gender	Group Category	Load	Speed (m/s)	Stride (m)	Cadence (spm)	Steps	Wait (s)	Origin Side	Destination Side	Transition
P1	M	Single	Load	0.01	0.11	11.6	5	16s	S-Kerb	S-Kerb	Curbside Wai
P2	M	Single	Load	0.02	0.13	20.3	31	184s	Median	Median	Median Wait
P4	F	Single	No L	0.02	0.11	16.4	4	14s	Median	Median	Median Wait
P5	M	Single	Load	0.01	0.10	7.6	6	157s	S-Kerb	S-Kerb	Curbside Wai
P6	F	Group	Load	0.01	0.10	13.0	132	692s	S-Kerb	S-Kerb	Curbside Wai
P7	M	Group	Load	0.02	0.12	18.4	97	447s	S-Kerb	S-Kerb	Curbside Wai
P9	M	Single	No L	0.01	0.10	8.0	2	13s	Median	Median	Median Wait
P13	M	Group	Load	0.03	0.13	19.6	4	18s	S-Kerb	S-Kerb	Curbside Wai
P14	M	Single	Load	0.01	0.10	6.3	6	74s	N-Kerb	N-Kerb	Curbside Wai
P15	F	Single	Load	0.00	0.10	2.5	2	40s	S-Kerb	S-Kerb	Curbside Wai
P16	M	Single	Load	0.01	0.11	11.8	14	79s	Median	Median	Median Wait
P17	F	Single	Load	0.12	0.32	37.2	10	1s	Median	Median	Median Wait
P18	M	Single	Load	0.06	0.21	34.4	2	4s	S-Kerb	S-Kerb	Curbside Wai
P20	M	Single	No L	0.01	0.10	11.4	1	4s	Median	Median	Median Wait
P21	M	Single	Load	0.01	0.10	8.1	4	54s	S-Kerb	S-Kerb	Curbside Wai
P24	M	Single	Load	0.00	0.10	5.2	1	45s	N-Kerb	N-Kerb	Curbside Wai
P25	M	Single	No L	0.02	0.11	24.7	2	4s	Median	Median	Median Wait
P26	M	Single	Load	0.01	0.10	10.0	3	20s	S-Kerb	S-Kerb	Curbside Wai
P29	F	Single	Load	0.03	0.13	25.3	3	4s	N-Kerb	N-Kerb	Curbside Wai
P33	F	Single	No L	0.02	0.10	18.6	1	4s	Median	Median	Median Wait
P35	M	Group	Load	0.02	0.13	17.4	68	389s	S-Kerb	S-Kerb	Curbside Wai
P37	F	Single	Load	0.03	0.12	26.5	5	9s	N-Kerb	N-Kerb	Curbside Wai
P38	M	Single	No L	0.03	0.13	27.9	2	5s	Median	Median	Median Wait
P41	M	Single	No L	0.01	0.10	11.4	2	9s	Median	Median	Median Wait
P42	M	Single	Load	0.04	0.15	25.8	27	130s	Median	Median	Corridor Cro
P43	F	Single	No L	0.13	0.41	38.6	2	1s	Median	Median	Median Wait
P45	F	Single	Load	0.01	0.11	15.1	77	259s	N-Kerb	N-Kerb	Curbside Wai
P46	M	Single	Load	0.05	0.20	28.9	5	7s	Median	Median	Median Wait
P47	M	Single	No L	0.01	0.10	10.8	3	13s	Median	Median	Median Wait
P48	M	Single	No L	0.03	0.13	23.5	3	11s	Median	Median	Median Wait
P49	F	Single	Load	0.04	0.17	29.9	5	9s	S-Kerb	S-Kerb	Curbside Wai
P50	F	Single	Load	0.08	0.25	36.2	31	99s	Median	Median	Corridor Cro

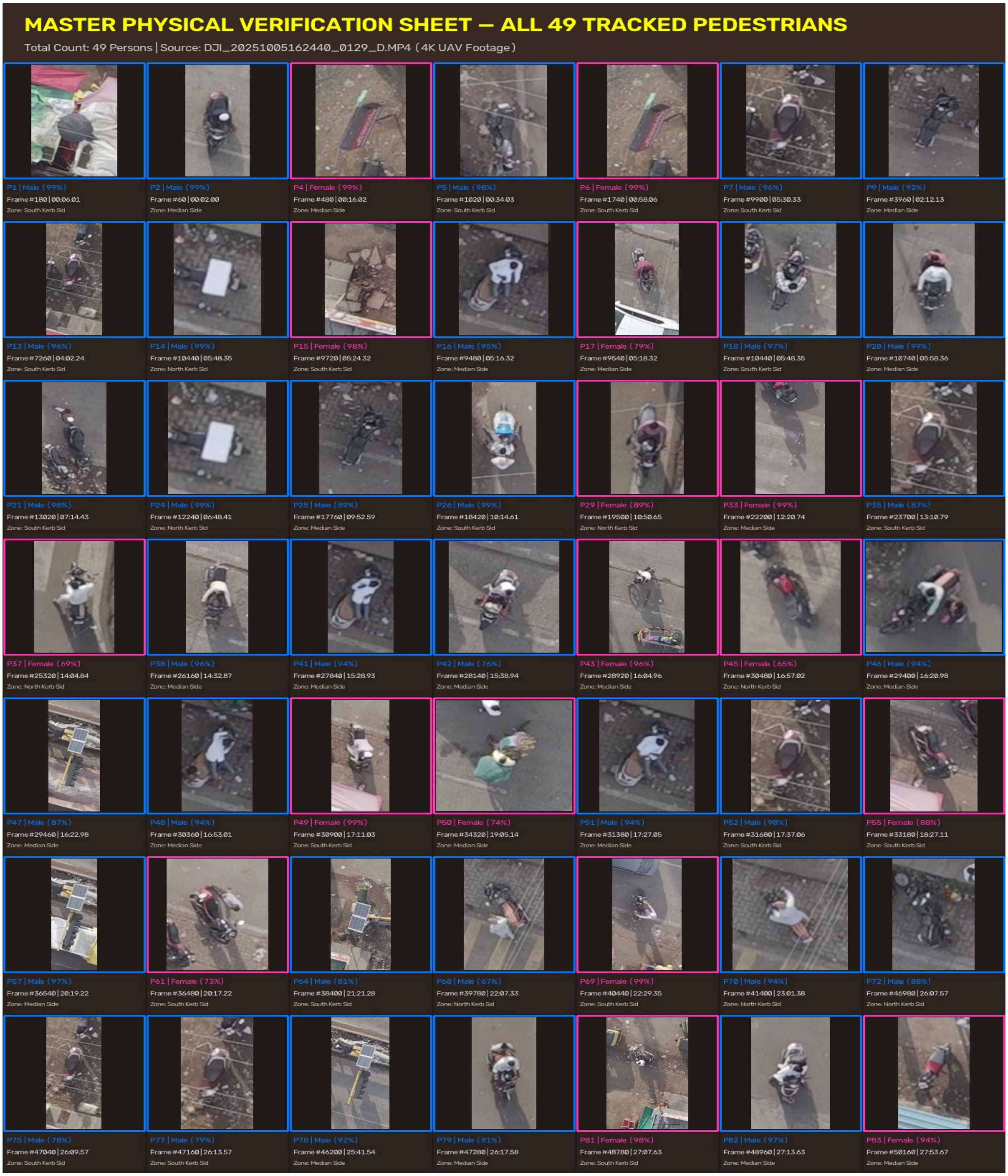
PID	Gender	Group Category	Load	Speed (m/s)	Stride (m)	Cadence (spm)	Steps	Wait (s)	Origin Side	Destination Side	Transition
P51	M	Single	No L	0.02	0.11	21.4	3	7s	Median	Median	Median Wait
P52	M	Single	No L	0.01	0.10	9.8	1	9s	S-Kerb	S-Kerb	Curbside Wai
P55	F	Single	Load	0.01	0.10	11.4	9	61s	S-Kerb	S-Kerb	Curbside Wai
P57	M	Single	Load	0.03	0.14	16.5	16	97s	Median	Median	Median Wait
P61	F	Single	No L	0.03	0.12	24.7	3	5s	S-Kerb	S-Kerb	Curbside Wai
P64	M	Single	Load	0.00	0.10	1.8	1	164s	S-Kerb	S-Kerb	Curbside Wai
P68	M	Single	Load	0.03	0.14	21.0	5	7s	N-Kerb	N-Kerb	Curbside Wai
P69	F	Single	Load	0.04	0.15	28.0	22	52s	S-Kerb	S-Kerb	Curbside Wai
P70	M	Single	Load	0.03	0.14	19.7	7	16s	N-Kerb	N-Kerb	Curbside Wai
P72	M	Single	Load	0.01	0.10	8.0	19	204s	N-Kerb	N-Kerb	Curbside Wai
P75	M	Couple	Load	0.03	0.15	20.4	17	151s	S-Kerb	S-Kerb	Curbside Wai
P77	M	Couple	Load	0.01	0.10	13.6	14	63s	S-Kerb	S-Kerb	Curbside Wai
P78	M	Single	Load	0.01	0.11	10.8	5	65s	Median	Median	Median Wait
P79	M	Single	Load	0.01	0.10	14.3	7	29s	Median	Median	Median Wait
P81	F	Single	Load	0.02	0.11	15.8	2	34s	S-Kerb	S-Kerb	Curbside Wai
P82	M	Single	Load	0.02	0.12	18.0	6	9s	Median	Median	Median Wait
P83	F	Single	No L	0.02	0.13	19.9	2	7s	Median	Median	Median Wait

5. PHYSICAL FRAME VERIFICATION & GROUND TRUTH EVIDENCE

To establish empirical verifiability against the original 4K UAV video (DJI_20251005162440_0129_D.MP4), every single detected identity was mapped back to its precise video timestamp, frame index, and spatial coordinates. A dedicated verification card was rendered for all 49 pedestrians, displaying the full video frame with targeting reticle and the high-resolution crop side-by-side.

Figure 5: Exemplary Physical Verification Card (Person #50, Female active corridor crosser, Frame #34320, t = 19m 05s)

Figure 6: Master Verification Mosaic Sheet — All 49 Tracked Pedestrians with Video Frame Numbers & Timestamps



6. DELIVERABLES DIRECTORY & SYSTEM FILE LOCATIONS

The exact physical locations of all generated datasets, models, visualizations, and video files on the local Windows filesystem:

Deliverable Name	Absolute Windows File Path
PDF Comprehensive Report	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\PEDESTRIAN_BEHAVIOR_AND_GAIT_MATHEMATICAL_REPORT.pdf
Physical Verification Index CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\pedestrian_physical_verification_index.csv
Physical Verification 49 Cards Folder	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\cards
Master Verification Mosaic Sheet	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\ALL_49_PERSONS_VERIFICATION_SHEET.jpg
Interactive HTML Verification Viewer	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\physical_verification_viewer.html
Per-Person Behavioral Features CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.csv
Per-Person Behavioral Features JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.json
Timeseries Kinematic CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_timeseries.csv
Gait & Crossing Summary CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.csv
Gait & Crossing Summary JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.json
Gait Timeseries CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_timeseries.csv
Perfected Zones JSON Specification	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\crossing_zones_perfect.json
Perfected Crossing Zones Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\perfect_crossing_zones_annotated.jpg
Trajectories Overlay Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_crossing_trajectories_annotated.jpg
8-Feature Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_features_dashboard.png
Gait & Crossing Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\gait_and_crossing_dashboard.png
Kerb vs. Median Crossing Spatial Map	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\kerb_median_crossing_map.jpg
Annotated 1080p Video (MP4, 26 MB)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\annotated_intersection_video.mp4
Physical Verification Extractor (Stage 19)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage19_extract_physical_verification_frames.py
Gait Dynamics Script (Stage 18)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage18_gait_and_crossing_dynamics.py
Features Extraction Script (Stage 16)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage16_pedestrian_comprehensive_features.py
Video Annotation Script (Stage 17)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage17_annotate_full_video.py
Refined Zones Script	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\refine_crossing_and_roadway.py